

RUIDONG ZHANG

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Education

University of Science and Technology of China(USTC)

Master of Science in Computer Science

Sep. 2025 – Present

Hefei, P.R.China

University of Electronic Science and Technology of China(UESTC)

Bachelor of Science in Computer Science(GPA: 3.84/4.0)

Sep. 2021 – Jun. 2025

Chengdu, P.R.China

Research Experience

Shanghai Artificial Intelligence Laboratory

Position: Research Intern

Jan. 2026 – Jul. 2026

Shanghai, P.R.China

- Designed and implemented an MLIR-based domain-specific compiler for GPU-intensive graphics workloads, preserving high-level application semantics through optimization and CUDA lowering.
- Developed semantics-guided compiler optimizations that coordinate program rewriting, computation placement, and GPU memory decisions across the compilation pipeline.
- Evaluated the compiler across representative workloads and multiple GPU architectures against hand-written CUDA and existing high-level programming systems, demonstrating substantial performance gains while improving programmability.

Industry Experience

Ubiquant Technology Co.

Position: Quantitative Implementation Intern

Aug. 2026 – Sep. 2026

Shanghai, P.R.China

- Evaluated multi-node, multi-GPU LLM serving with SGLang and vLLM under prefill-decoding disaggregation, characterizing TTFT/TPOT trade-offs across distributed execution configurations.
- Investigated Mooncake-based KV-cache transfer and profiled compute, memory, and communication bottlenecks across prefill/decode placement strategies.

Houmo Technology Co.

Position: AI Compiler Development Intern

Apr. 2025 – Jul. 2025

Nanjing, P.R.China

- Implemented 44 ONNX operators end-to-end for a RISC-V-based compute-in-memory NPU, spanning MLIR dialect definitions, pattern rewriting, and lowering to accelerator-specific primitives.
- Built MLIR lowering and validation pipelines that mapped ONNX operators to backend execution constraints and regression-tested outputs against ONNX Runtime.

Projects

Compiler-Assisted Execution for Extensible Block-Diagram Simulation

Jun. 2026 - Present

- Designed a compiler-assisted execution framework that lowers dynamically dispatched block-diagram models into compiled execution regions while preserving solver and state-update semantics.
- Developed dependency-aware specialization to avoid redundant model evaluation during multi-stage ODE integration, compiling only computations affected by intermediate solver states.
- Built a JIT execution path with cached native code and runtime fallback, supporting incremental compilation of eligible operator regions without replacing the existing simulation runtime.

RISC-V Processor Design & Architectural Validation

Apr. 2024 - Oct. 2025

- Designed a five-stage RV32E/RV32I processor with forwarding/hazard control, branch handling, CSR exceptions/interrupts, and an AXI4 memory subsystem; deployed it on an Artix-7 FPGA and booted RT-Thread.
- Built an RV32IM emulator with breakpoints, disassembly, and device simulation, together with Spike differential testing for ISA-level validation.

Awards

University of Science and Technology of China

First-Class Academic Scholarship

HP Responsible Entrepreneurship Competition

First Prize

IEEE Extreme Programming Competition

Top 5% Worldwide

Technical Skills

Programming: C++, C, Python, CUDA, Verilog

Compiler: MLIR, LLVM

GPU & AI Systems: CUDA, ONNX, PyTorch, SGLang, vLLM